

Core Python

- Introduction to Python
- What is Python?
- History of Python
- Python Versions
- Features of Python
- How to Install Python
- Install Python with Diff IDEs
- Creating Your First Python Program
- Printing to the Screen
- Reading Keyboard Input
- Using Command Prompt and GUI or IDE **Different Modes in Python**
- Execute the Script
- Interactive Mode
- Script Mode
- Python Comments
- Working with Python in Unix/Linux/Windows/Mac/Android
- Python New IDEs
- What is PIP?

Variables in Python

- What is Variable?
- Variables in Python
- Constants in Python
- Standard Data Types
- Operators and Operands
- Swap variables
- Type Conversion
- String Handling

Python Conditional Statements

- How to use “if condition” in conditional structures □ if statement (One-Way Decisions)
- if .. else statement (Two-way Decisions)
- How to use “else condition”
- if ..elif .. else statement (Multi-way)
- When “else condition” does not work
- How to use “elif” condition
- How to execute conditional statement with minimal code
- Nested IF Statement

Python LOOPS

- How to use “While Loop”
- How to use “For Loop”
- How to use For Loop for set of other things besides numbers
- Break statements in For Loop
- Continue statement in For Loop
- Enumerate function for For Loop

Python Lists

- Lists are mutable
- Getting to Lists
- List indices
- Traversing a list
- List operations
- List slices
- List methods
- Map, filter and reduce

Python TUPLE

- Advantages of Tuple over List
- Packing and Unpacking
- Comparing tuples
- Creating nested tuple
- Using tuples as keys in dictionaries
- Deleting Tuples
- Slicing of Tuple
- Tuple Membership Test

Python Sets

- How to create a set?
- Iteration Over Sets
- Python Set Methods
- Python Set Operations
- Union of sets
- Built-in Functions with Set
- Python Frozenset

Python Dictionary

- How to create a dictionary? □ Python Hashing?
- Python Dictionary Methods
- Copying dictionary
- Updating Dictionary
- Delete Keys from the dictionary
- Dictionary items() Method
- Sorting the Dictionary
- Python Dictionary in-built Functions

Python Functions

- What is a Function?
- How to define and call a function in Python
- Types of Functions
- Significance of Indentation (Space) in Python □ How Function Return Value?
- Types of Arguments in Functions
- Default Arguments
- Non-Default Arguments

- Keyword Arguments
- Non-keyword Arguments
- Arbitrary Arguments
- Rules to define a function in Python
- Various Forms of Function Arguments
- Scope and Lifetime of variables
- Anonymous Functions/Lambda functions
- map(), filter(), reduce() functions □ What is a Docstring?

Advanced Python

- Python Exception Handling
- Python Errors
- Common RunTime Errors in PYTHON
- Abnormal termination
- Chain of importance Of Exception
- Exception Handling
- Try ... Except
- Try ..Except .. else
- Try ... finally

Python Class and Objects

- Introduction to OOPs Programming
- Object Oriented Programming System
- OOPS Principles
- Basic concept of Object and Classes
- Access Modifiers
- How to define Python classes
- Self-variable in python
- What is Inheritance? Types of Inheritance?
- How Inheritance works?

Python Regular Expressions

- What is Regular Expression?
- Regular Expression Syntax
- Understanding Regular Expressions
- Regular Expression Patterns
- Literal characters